

Report PDF Compilation Rule Update

Date: 2026-07-05

Request

The user requested that every answer written into report/ should be automatically compiled to PDF and that this workflow should be recorded in `.ai_rules`.

Rule Added

`.ai_rules` now requires:

- every Markdown report in report/ to be compiled into a same-name PDF;
- the local compiler `python report/compile_report_pdf.py <report.md>` to be used by default;
- pandoc with xelatex to be acceptable if available;
- the task not to be considered complete until PDF generation has been attempted and reported;
- missing PDF tooling to be treated as a blocker unless it can be installed or configured.

Current Toolchain Check

Checked tools in the current environment:

- pandoc: not available
- xelatex: not available
- wkhtmltopdf: not available
- weasyprint: installed successfully through `python -m pip install weasyprint`
- latex2mathml: installed successfully through `python -m pip install latex2mathml`
- Python markdown: available
- Python reportlab: not available

Status

The rule has been written and a local compiler has been added:

```
python report/compile_report_pdf.py
```

All current Markdown reports in report/ have been compiled to same-name PDFs:

- README.pdf
- mab_model_algorithm_review.pdf
- pdf_compile_rule_update.pdf
- tpot_cost_reward_update.pdf